

Quasi-Poisson models

Last time: modeling article publication

We are interested in analyzing the number of articles published by biochemistry PhD students. The data contains the following variables:

- + art: articles published in last three years of Ph.D.
- + fem: gender (recorded as male or female)
- + mar: marital status (recorded as married or single)
- + kid5: number of children under age six
- + phd: prestige of Ph.D. program
- + ment: articles published by their mentor in last three years

$$Articles_i \sim Poisson(\lambda_i)$$

$$\log(\lambda_i) = \beta_0 + \beta_1 Female_i + \beta_2 Married_i + \beta_3 Kids_i + \beta_4 Prestige_i + \beta_5 Mentor_i$$

Recap: confidence interval

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	0.304617	0.102981	2.958	0.0031	**
femWomen	-0.224594	0.054613	-4.112	3.92e-05	***
marMarried	0.155243	0.061374	2.529	0.0114	*
kid5	-0.184883	0.040127	-4.607	4.08e-06	***
phd	0.012823	0.026397	0.486	0.6271	
ment	0.025543	0.002006	12.733	< 2e-16	***

Confidence interval for β_4 :

Goodness of fit test

Null deviance: 1817.4 on 914 degrees of freedom
Residual deviance: 1634.4 on 909 degrees of freedom

Offset?

- + art: articles published in last three years of Ph.D.
- + fem: gender (recorded as male or female)
- + mar: marital status (recorded as married or single)
- + kid5: number of children under age six
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$$Articles_i \sim Poisson(\lambda_i)$$

$$\log(\lambda_i) = \beta_0 + \beta_1 Female_i + \beta_2 Married_i + \beta_3 Kids_i + \beta_4 Prestige_i + \beta_5 Mentor_i$$

Do you think this model needs an offset?

Overdispersion

Overdispersion occurs when the response Y has higher variance than we would expect if Y followed a Poisson distribution.

Formally, let

$$\phi = \frac{\text{Variance}}{\text{Mean}}$$

What should ϕ be if there is no overdispersion?

Estimating overdispersion

The *Pearson residual* for observation i is defined as

$$e_{(P)i} = \frac{Y_i - \hat{\lambda}_i}{\sqrt{\hat{\lambda}_i}}$$

$$\hat{\phi} = \frac{\sum_{i=1}^n e_{(P)i}^2}{n - p}$$

+ p = number of parameters in model

Example: Estimating overdispersion

```
# fit the model  
m1 <- glm(art ~ ., data = articles,  
          family = poisson)  
  
# get Pearson residuals  
pearson_resids <- resid(m1, "pearson")  
  
# estimate dispersion parameter  
phihat <- sum(pearson_resids^2)/(915 - 6)  
phihat
```

```
## [1] 1.828984
```


Handling overdispersion

Overdispersion is a problem because our standard errors (for confidence intervals and hypothesis tests) are too low.

If we think there is overdispersion, what should we do?

Adjusting the standard error

- + In our data, $\hat{\phi} = 1.83$
- + This means our variance is 1.83 times bigger than it should be
- + So our standard error is $\sqrt{1.83} = 1.35$ times bigger than it should be

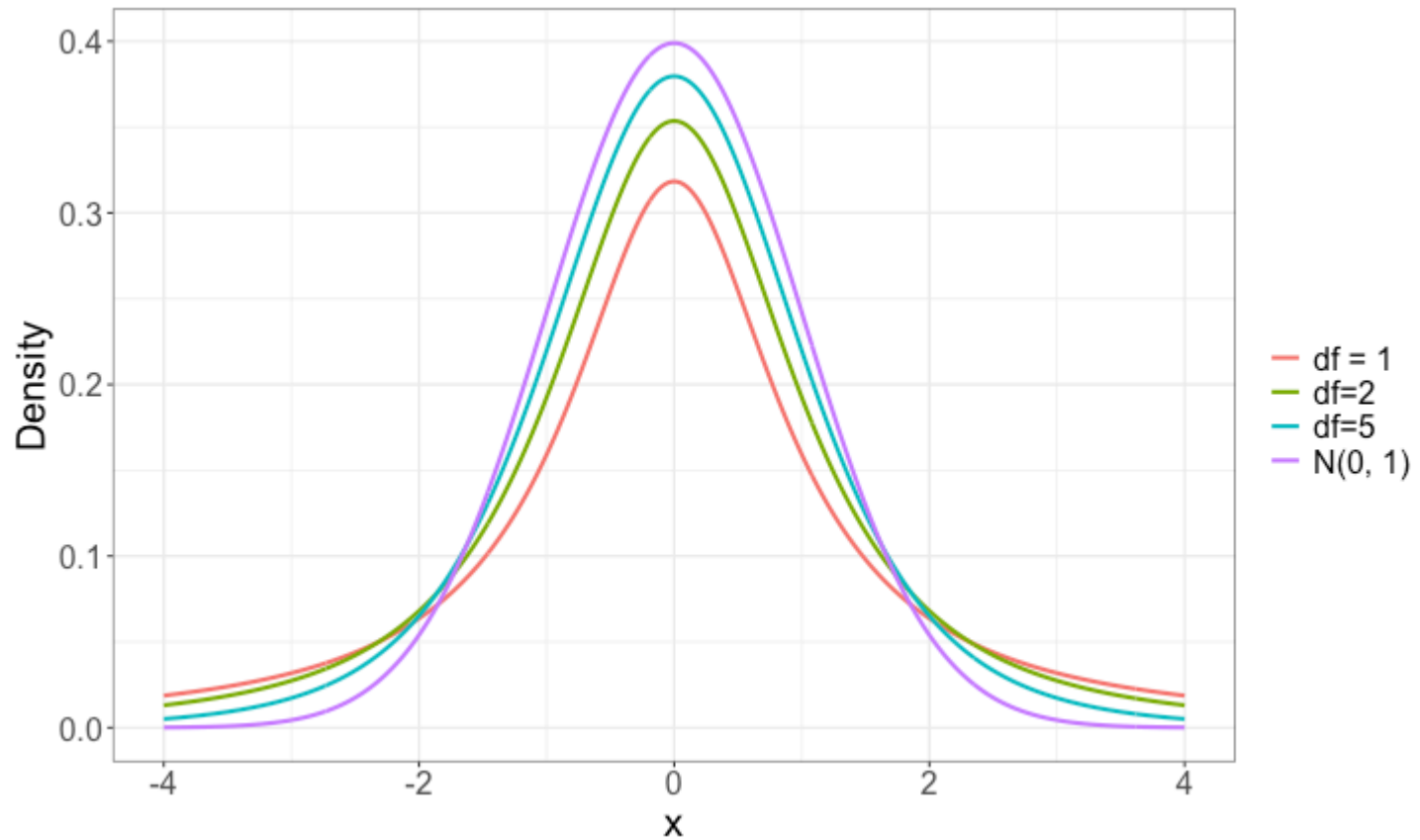
Adjusting the standard error in R

```
m2 <- glm(art ~ ., data = articles,  
          family = quasipoisson)
```

```
...  
## kid5          -0.184883    0.054268   -3.407 0.000686 ***  
## phd           0.012823    0.035700    0.359 0.719544  
## ment          0.025543    0.002713    9.415 < 2e-16 ***  
## ---  
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1  
##  
## (Dispersion parameter for quasipoisson family taken to be 1.829006)  
##  
##      Null deviance: 1817.4  on 914  degrees of freedom  
## Residual deviance: 1634.4  on 909  degrees of freedom  
## AIC: NA  
...
```

- ✚ Allowing ϕ to be different from 1 means we are using a *quasi-likelihood* (in this case, a *quasi-Poisson*)

t-distribution



Calculating a confidence interval

```
...  
## phd          0.012823    0.035700    0.359 0.719544  
## ment        0.025543    0.002713    9.415 < 2e-16 ***  
## ---  
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1  
##  
## (Dispersion parameter for quasipoisson family taken to be 1.829006)  
...
```

New confidence interval for β_4 :

$$0.0128 \pm t_{n-p}^* \cdot 0.0357$$

```
qt(0.025, df=909, lower.tail=F)
```

```
## [1] 1.962577
```

$$0.0128 \pm 1.96 \cdot 0.0357 = (-0.0572, 0.0828)$$

Adjusting the standard error in R

Poisson:

```
...  
## kid5          -0.184883    0.040127   -4.607  4.08e-06 ***  
## phd           0.012823    0.026397    0.486   0.6271  
## ment          0.025543    0.002006   12.733  < 2e-16 ***  
## ---  
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1  
...
```

Quasi-Poisson:

```
...  
## kid5          -0.184883    0.054268   -3.407  0.000686 ***  
## phd           0.012823    0.035700    0.359  0.719544  
## ment          0.025543    0.002713    9.415  < 2e-16 ***  
## ---  
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1  
...
```

Class activity

Work on the class activity. Submit your rendered HTML on Canvas at the end of class.